SMART INDIA HACKATHON 2025

TITLE PAGE

- **Problem Statement ID –** 25050
- **Problem Statement Title-** Smart Traffic Management system for urban congestion
- **Theme-** Transportation and logistics
- **PS Category-** Software
- **Team Name-** Team DUI

Team DUI

Solving traffic Using MARL models

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Uniqueness of our solution

•Builds on **QMIX** but extends it with richer real-world traffic data integration.
•Leverages **MARL cooperation**, enabling global optimization beyond local decisions.
•**Data-driven** approach ensures adaptability across cities and conditions.
•Outperforms traditional systems in efficiency, coordination, and sustainability.
•Grounded in **latest MARL research**, while moving beyond simulation to practical deployment.

A Custom Dashboard And A Sim

- A user-friendly dashboard displays real-time traffic metrics, agent decisions, and performance statistics, allowing city authorities to monitor, analyze, and fine-tune traffic control strategies effectively.
- This also allows the user to do manage traffic hands on, providing a safe guard against any malfunctions
- SUMO is a very helpful tool which allows a lot of customizability.

Proposed Solution

•Using **Multi-Agent Reinforcement Learning (MARL)** to control traffic lights, where each signal acts as an intelligent agent.
•Start with **QMIX** as the baseline algorithm for cooperative learning.
•Integrate **traffic data from OpenCV** along with contextual features (time of day, road layout, etc.) to guide learning.
•The baseline model was trained in a **SUMO simulation** environment to achieve a reliable level of performance.

Addressing The Problem

•**Adapts in real-time**: Unlike fixed timers (like in Odisha), our system actually learns and adjusts to current traffic instead of running on a set schedule.
•**Cuts congestion**: By letting traffic lights “talk” to each other, intersections don’t get stuck and long queues are avoided.
•**Saves time & fuel**: Less waiting at red lights means smoother flow, lower fuel use, and fewer emissions.

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TECHNICAL APPROACH

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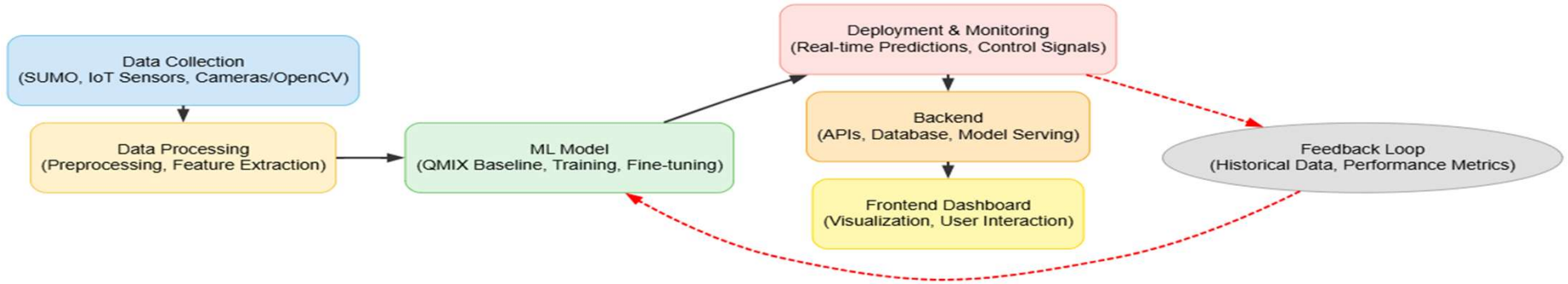
Software

- **Front End:** HTML, JavaScript, CSS
- **Back End:** Python (Flask)
- **Simulator:** SUMO
- **AI/ML Model Code:** Python
- **Connector:** TraCI API
- **Module for Traffic Volume:** Python (OpenCV)

Hardware

IOT sensors, Cameras, Traffic lights

Process Flow Architecture



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FEASIBILITY AND VIABILITY

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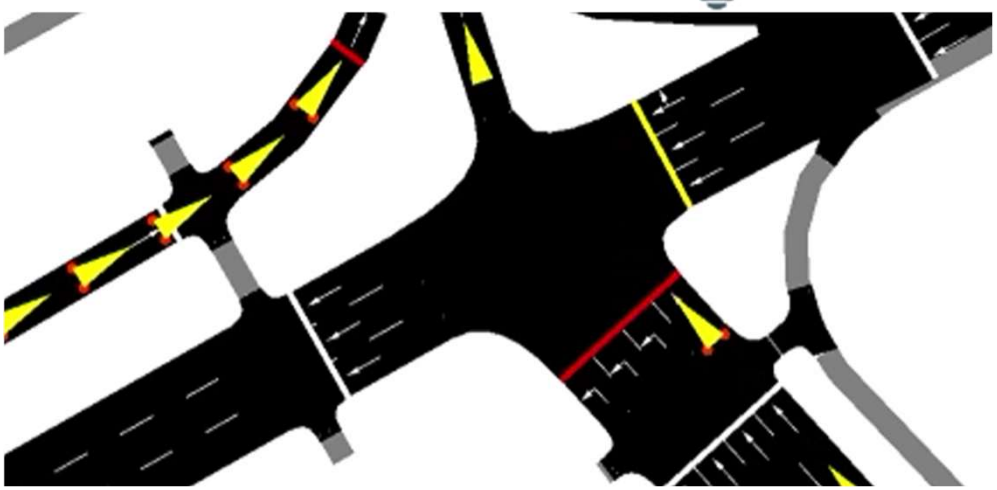
Why is it feasible?

•**Proven approach**: MARL with QMIX has shown success in multi-agent coordination, making it suitable for traffic optimization.
•**Simulation support**: SUMO provides a realistic environment for training and benchmarking models.
•**Scalable design**: Centralized training and decentralized execution allow stepwise deployment, from single intersections to citywide networks.
•**Impact**: Long-term benefits include reduced congestion, fuel savings, lower emissions, and improved travel times.



Overcoming the challenges

•**Domain adaptation & transfer learning**: Bridge the gap between SUMO simulations and real-world deployment.
•**Advanced MARL methods**: Use hierarchical or graph-based MARL to manage large-scale road networks efficiently.
•**Data integration**: Combine OpenCV datasets with sensor, GPS, and crowdsourced mobility data.
•**Incremental deployment**: Start with hybrid AI-assisted systems in areas with limited infrastructure.
•**QMIX Variation-** Using a different variation of QMIX like QTRAN++ can improve scalability of the model.

Potential Challenges and risks

•**Simulation-to-reality gap**: Models trained in SUMO may not directly translate to real-world conditions.
•**Scalability**: Coordinating a large number of intersections increases complexity and is hard.
•**Data limitations**: Real-time traffic data may be inconsistent or unavailable in some areas.
•**Infrastructure readiness**: Not all cities have the digital systems needed for MARL-based control.
•**In India**: A QMIX model hasn't been used within India's context, so it may lead to some very glaring issues

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IMPACT AND BENEFITS

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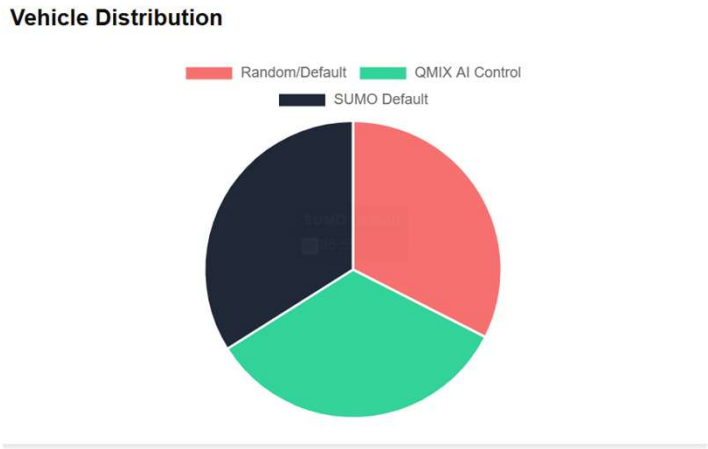
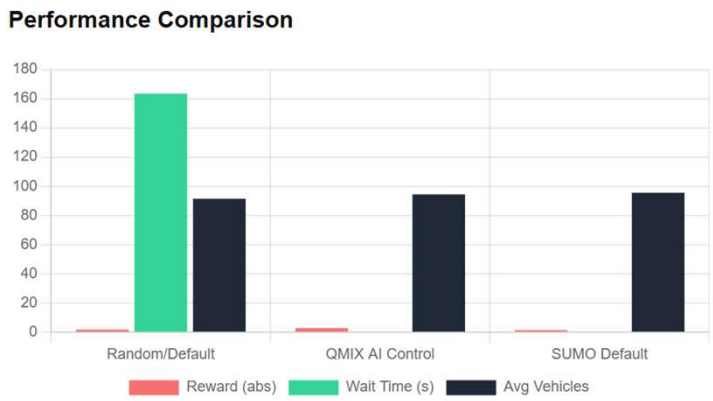
Potential Impact on the Target Audience

•**Commuters**: Reduced waiting times at traffic lights, smoother journeys, and less daily stress.
•**Public transport**: Buses and other services experience fewer delays, improving reliability.
•**Emergency services**: Ambulances, fire trucks, and other priority vehicles can navigate traffic more efficiently.
•**City authorities**: Provides a data-driven system that simplifies traffic management and planning.
•**General public**: Cleaner air, lower noise levels, and more organized traffic flow.





Benefits of the Solution

•**Social**: Safer streets, smoother traffic flow, and improved pedestrian experiences.
•**Economic**: Less fuel consumption, shorter travel times, and increased overall productivity.
•**Environmental**: Reduced vehicle emissions and lower pollution from idling traffic.
•**Technological/Future-ready**: System can integrate with IoT devices and smart city infrastructure as they become available.

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RESEARCH AND REFERENCES

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- https://dl.acm.org/doi/abs/10.5555/3455716.3455894
- https://ieeexplore.ieee.org/document/10887881/
- https://arxiv.org/abs/1803.11485
- https://www.researchgate.net/publication/349020721_Multiple_agents_cooperative_control_based_on_QMIX_algorithm_in_SC2LE_environment
- https://www.youtube.com/watch?v=mAkGYR0ykX4